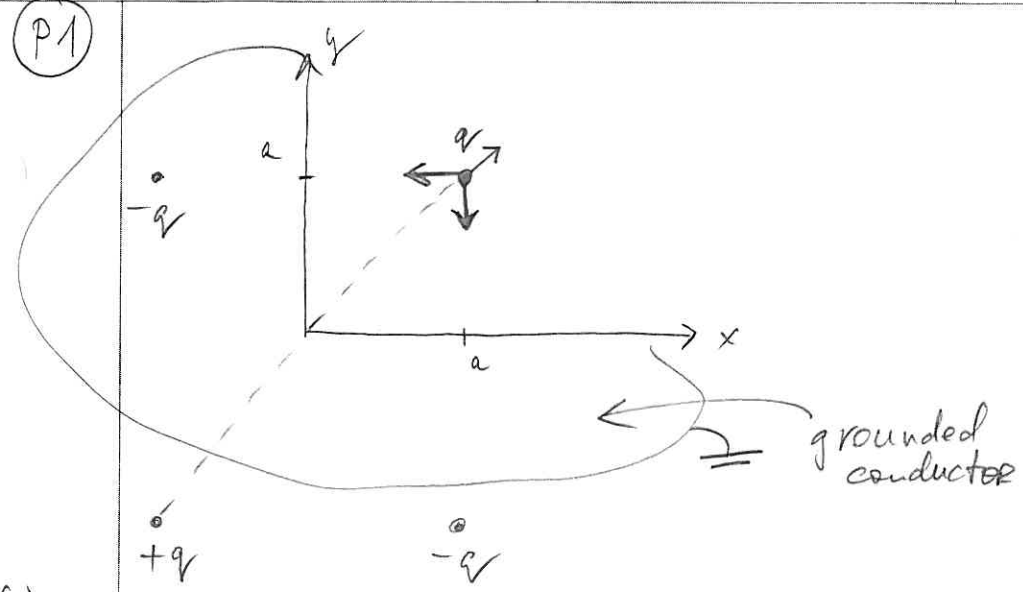
Midterm II Solutions

P1



(a)

$$x>0, y>0: \quad \phi = \frac{q}{4\pi\varepsilon_0}\left\{\frac{1}{\sqrt{(x-a)^2+(y-a)^2}} + \frac{1}{\sqrt{(x+a)^2+(y+a)^2}} - \frac{1}{\sqrt{(x-a)^2+(y+a)^2}} - \frac{1}{\sqrt{(x+a)^2+(y-a)^2}}\right\}$$

Setting $x=0$ or $y=0$ gives $\phi=0$, as it should be.

(b) Force on q is produced by 3 image charges:

$$\vec{F}_q = \frac{-q^2}{4\pi\varepsilon_0}\left\{\frac{\hat{y}}{(2a)^2} + \frac{\hat{x}}{(2a)^2} - \frac{\cos\frac{\pi}{4}\hat{x} + \sin\frac{\pi}{4}\hat{y}}{(4a^2+4a^2)}\right\}$$

$$= \frac{-q^2}{4\pi\varepsilon_0}\frac{1}{4a^2}\left(1-\frac{1}{2\sqrt{2}}\right)(\hat{x}+\hat{y})$$ attraction along diagonal direction.

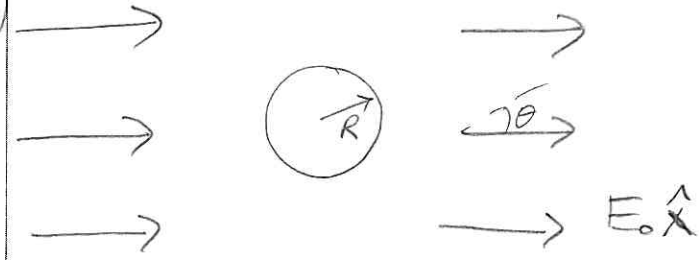
(c) Work is determined by electrostatic energy U

$W = \frac{1}{4}U$ but only $\frac{1}{4}$ because $\phi=0$ ($\vec{E}=0$) in 3 quarters of the space!

$$U = \frac{q^2}{4\pi\varepsilon_0}\left(\frac{1}{2a} + \frac{1}{2a} - \frac{1}{2\sqrt{2}a}\right)$$

The same can be obtained by integrating $\vec{F}_q$ from (b) along the diagonal path $\vec{r}=(p,p)$ with p from a to ∞.

P2



$\varepsilon_r > 1$

$x = r\cos\theta$

Inside the buble field is uniform $\Rightarrow$ electric potential

<u>$r<R$</u> $\quad \phi_{in} = \gamma r\cos\theta$ (so that $\vec{E}_{in} = -\gamma\hat{x}$)

<u>$r>R$</u> $\quad \phi_{out} = -E_0 r\cos\theta + \dfrac{\beta\cos\theta}{r^2}$

(so that for from buble $\vec{E} = -\vec{\nabla}\phi_{out} = E_0\hat{x}$)

<u>BC</u> $\quad D_{out,\perp} = D_{in,\perp} \Rightarrow \varepsilon_0\varepsilon_r \left.\dfrac{\partial\phi_{out}}{\partial r}\right|_R = \varepsilon_0 \left.\dfrac{\partial\phi_{in}}{\partial r}\right|_R \quad (1)$

$E_{out,\parallel} = E_{in,\parallel} \Rightarrow \dfrac{1}{R}\dfrac{\partial\phi_{out}}{\partial\theta} = \dfrac{1}{R}\dfrac{\partial\phi_{in}}{\partial\theta} \quad (2)$

Thus (1): $\quad \varepsilon_r\left(-E_0\cos\theta - 2\dfrac{\beta\cos\theta}{R^3}\right) = \gamma\cos\theta$

$$\Rightarrow E_0 + \frac{2\beta}{R^3} = -\frac{\gamma}{\varepsilon_r}$$

(2) $\quad E_0 R\sin\theta - \dfrac{\beta\sin\theta}{R^2} = -\gamma R\sin\theta$

$$\Rightarrow E_0 - \frac{\beta}{R^3} = -\gamma$$

$$\Rightarrow \frac{3\beta}{R^3} = -\frac{\gamma}{\varepsilon_r} + \gamma = \gamma\left(1 - \frac{1}{\varepsilon_r}\right) = \gamma\,\frac{\varepsilon_r - 1}{\varepsilon_r}$$

$$\gamma = -E_0 + \frac{\beta}{R^3} = -E_0 + \gamma\,\frac{\varepsilon_r - 1}{3\varepsilon_r}$$

$$\gamma\left(1 - \frac{\varepsilon_r - 1}{3\varepsilon_r}\right) = -E_0, \quad \gamma\,\frac{2\varepsilon_r + 1}{3\varepsilon_r} = -E_0$$

$$\Rightarrow \vec{E}_{in} = \frac{3\varepsilon_r}{2\varepsilon_r + 1}E_0\hat{x}.$$

(P3) Current along ABCD produces $B_0\hat{y}$.

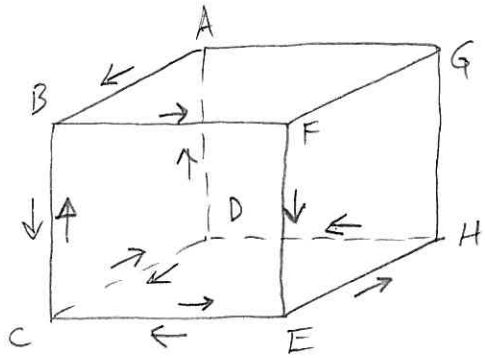
$\Rightarrow$ Current along BFEC produces $B_0(-\hat{x})$.

Current along EHDC produces $B_0(\hat{z})$.

Note that currents on lines BC, CE, FG, DC cancel each other resulting in pattern in Fig 2(b).

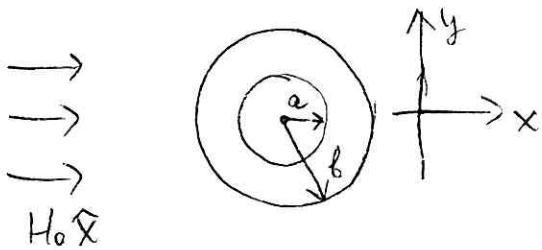
Thus the net field at the center of the cube is

$$\vec{B} = B_0\left(-\hat{x} + \hat{y} + \hat{z}\right), \quad |\vec{B}| = \sqrt{3}\, B_0$$



Superposition gives Fig. 2b!

P4



$$\left.\begin{aligned}\vec{\nabla}\times\vec{H}&=0\\ \vec{\nabla}\cdot\vec{B}&=0\end{aligned}\right\}\quad \text{let } \vec{B}=-\vec{\nabla}\phi_M,\quad \nabla^2\phi_M=0$$

with boundry conditions $H_{\parallel}$, $B_{\perp}$ are continuous at $\rho=a$, $\rho=b$.

General solution

$$\phi_M(\rho,\varphi)=a_0+b_0\ln\rho+\sum_{n=1}^{\infty}\left(a_n\rho^n+\frac{b_n}{\rho^n}\right)\cos(n\varphi)$$

1) $\rho>b$: $\phi_M=-\mu_0 H_0\rho\cos\varphi+\frac{B_1}{\rho}\cos\varphi$

2) $a<\rho<b$: $\phi_M=+A_2\rho\cos\varphi+\frac{B_2}{\rho}\cos\varphi$

3) $\rho<a$: $\phi_M=A_3\rho\cos\varphi$ (must be regular at $\rho=0$)

$$\vec{B}=-\vec{\nabla}\phi_M=-\frac{\partial\phi_M}{\partial\rho}\hat{\rho}-\frac{1}{\rho}\frac{\partial\phi_M}{\partial\varphi}\hat{\varphi}$$

BC: $\underline{\rho=a-0}$ $\quad B_{\perp}(a-0)=B_{\rho}(a-0)=-A_3\cos\varphi$

$$B_{\parallel}=B_{\varphi}=A_3\sin\varphi$$

$\rho=a+0$: $\quad B_{\perp}=-A_2\cos\varphi+\frac{B_2}{a^2}\cos\varphi$

$$B_{\parallel}=A_2\sin\varphi+\frac{B_2}{a^2}\sin\varphi$$

$\rho=b-0$: $\quad B_{\perp}=-A_2\cos\varphi+\frac{B_2}{b^2}\cos\varphi$

$$B_{\parallel}=A_2\sin\varphi+\frac{B_2}{b^2}\sin\varphi$$

$\rho = b+0$: $\quad B_{\perp} = \mu_0 H_0 \cos\varphi + \frac{B_1}{b^2}\cos\varphi$

$$B_{\parallel} = -\mu_0 H_0 \sin\varphi + \frac{B_1}{b^2}\sin\varphi$$

$$\Rightarrow \quad -A_3 \cos\varphi = -A_2 \cos\varphi + \frac{B_2}{a^2}\cos\varphi$$

$$A_3 = A_2 - \frac{B_2}{a^2} \qquad (1)$$

$$\frac{1}{\mu_0} A_3 \sin\varphi = \frac{1}{\mu}\left(A_2 \sin\varphi + \frac{B_2}{a^2}\sin\varphi\right)$$

$$A_3 = \frac{\mu_0}{\mu}\left(A_2 + \frac{B_2}{a^2}\right) \qquad (2)$$

$$-A_2 \cos\varphi + \frac{B_2}{b^2}\cos\varphi = \mu_0 H_0 \cos\varphi + \frac{B_1}{b^2}\cos\varphi$$

$$A_2 - \frac{B_2}{b^2} = -\mu_0 H_0 - \frac{B_1}{b^2} \qquad (3)$$

$$\frac{1}{\mu}\left(A_2 \sin\varphi + \frac{B_2}{b^2}\sin\varphi\right) = \frac{1}{\mu_0}\left(-\mu_0 H_0 \sin\varphi + \frac{B_1}{b^2}\sin\varphi\right)$$

$$\frac{\mu_0}{\mu}\left(A_2 + \frac{B_2}{b^2}\right) = -\mu_0 H_0 + \frac{B_1}{b^2} \qquad (4)$$

$$A_2 + \frac{B_2}{b^2} = \frac{\mu}{\mu_0}\left(-\mu_0 H_0 + \frac{B_1}{b^2}\right) \qquad (4')$$

$$\frac{B_1}{b^2} = -\mu_0 H_0 - A_2 + \frac{B_2}{b^2} \quad (3')$$

$$A_2 + \frac{B_2}{b^2} = \frac{\mu}{\mu_0}\left(-2\mu_0 H_0 - A_2 + \frac{B_2}{b^2}\right) \quad (4')$$

$$\Rightarrow A_2\left(1+\frac{\mu}{\mu_0}\right) + \frac{B_2}{b^2}\left(1-\frac{\mu}{\mu_0}\right) = -2\mu H_0 \quad (4'')$$

(1)−(2) gives: $0 = \left(1-\frac{\mu_0}{\mu}\right)A_2 - \left(1+\frac{\mu_0}{\mu}\right)\frac{B_2}{a^2}$

$$(\mu+\mu_0)\,B_2 = a^2(\mu-\mu_0)\,A_2$$

let $\mu' = \frac{\mu}{\mu_0} \Rightarrow (\mu'+1)\,B_2 = a^2(\mu'-1)\,A_2$

Plug in (4''): $A_2(\mu'+1) - \frac{B_2}{b^2}(\mu'-1) = -2\mu'\mu_0 H_0$

$$\Rightarrow A_2\left[(\mu'+1) - \frac{a^2}{b^2}\frac{(\mu'-1)^2}{\mu'+1}\right] = -2\mu'\mu_0 H_0$$

$$A_2 = \frac{-2\mu'(\mu'+1)\mu_0 H_0}{(\mu'+1)^2 - \frac{a^2}{b^2}(\mu'-1)^2}$$

$$\Rightarrow B_2 = \frac{-2\mu'(\mu'-1)\,a^2\mu_0 H_0}{(\mu'+1)^2 - \frac{a^2}{b^2}(\mu'-1)^2}$$

Then $A_3 = A_2 - \frac{B_2}{a^2} = \frac{-2\mu'(\mu_0 H_0)\left[\mu'+1-(\mu'-1)\right]}{(\mu'+1)^2 - \frac{a^2}{b^2}(\mu'-1)^2} =$

$$= \frac{-4\mu'\mu_0 H_0}{(\mu'+1)^2 - \frac{a^2}{b^2}(\mu'-1)^2}$$

$$\Rightarrow \vec{B}(\rho<a) = -A_3\hat{x} =$$

$$= \frac{4\mu H_0}{(\mu'+1)^2 - \frac{a^2}{b^2}(\mu'-1)^2} \approx \frac{4H_0}{\mu'\left(1-\frac{a^2}{b^2}\right)} \ll \mu_0 H_0$$